TEXAS A&M UNIVERSITY

Proofs

CSCE 222: Discrete Structures for Computing

Rules of Inference

To prove mathematical statements, we must give valid arguments for them:

1. Start with statements we already know are true
2. Deduce new true statements from the ones we already know
3. Reach a conclusion

Rules of Inference:

The means for deducing new true statements.

Rules of Inference

e.g.

1. If you know your eCampus password, **then** you can see your grade.
2. You know your eCampus password.
3. Conclusion: You can see your grade.

If $\mathbf{p} \rightarrow \mathbf{q}$ is true and p is true, then q is true.

Rules of inference

Templates

Rules of Inference

Rules of inference have templates for constructing arguments.

1. Start with a collection of propositions, p_1, p_2,..., p_n: **the premises**
2. End with another proposition, q: **the conclusion**.

Let's review !

Rules of Inference

- Modus Ponens
- Modus Tollens
- Hypothetical syllogism
- Disjunctive syllogism
- Addition
- Simplification
- Conjunction
- Resolution

Modus Ponens

Modus ponens is the rule that lets you conclude something is true when you have a conditional statement and you know its premise is true.

* Latin for: ***mode that affirms***

The tautology $[p \land (p \rightarrow q)] \rightarrow q$ is the basis of the rule of inference.

p
$p \rightarrow q$

$\therefore q$

Modus Tollens

Modus tollens is the rule that lets you conclude the premise is false when you have a conditional statement and you know its conclusion is false.

* Latin for: **method of denying**

$\neg q$
$p \rightarrow q$

$\therefore \neg p$

$(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$

Hypothetical syllogism

Hypothetical syllogism is the rule that lets you chain together two conditional statements to create a new direct conditional relationship.

$p \rightarrow q$
$q \rightarrow r$

$\therefore p \rightarrow r$

$(p \rightarrow q \wedge q \rightarrow r) \rightarrow (p \rightarrow r)$

Disjunctive syllogism

Disjunctive syllogism is the rule that lets you conclude one alternative is true when you know at least one of two things must be true and one of them is false.

$p \vee q$
$\neg p$

$\therefore q$

$[(p \vee q) \wedge \neg p] \rightarrow q$

Addition

Addition is the rule that lets you add any alternative to a true statement to create a valid "or" statement.

p

∴ p ∨ q

$p \rightarrow p \vee q$

Simplification

Simplification is the rule that lets you extract a single component from a conjunction of multiple true statements.

$p \wedge q$

$\therefore p$

$p \wedge q \rightarrow p$

Conjunction

Conjunction is the rule that lets you combine two separately true statements into one compound statement.

p
q

∴ p ∧ q

$[(p) \wedge (q)] \rightarrow p \wedge q$

Resolution

Resolution is the rule that lets you combine two disjunctions containing opposite versions of the same component to eliminate that component and merge the remaining parts.

$p \vee q$
$\neg p \vee r$

$\therefore q \vee r$

$[(p \vee q) \wedge (\neg p \vee r)] \rightarrow q \vee r$

e.g.:

Either you know the password or you use your fingerprint.

Either you don't know password or you buy a new laptop.

∴ Either you use your fingerprint or you buy a new laptop.

Fallacies

Fallacy = error in reasoning that results in an **invalid argument**

Types:

- Converse Error
- Inverse Error

Converse Error

Converse Error is the logical fallacy of incorrectly concluding that the premise must be true simply because the conclusion of a conditional statement is true.

$p \rightarrow q$
q

$\therefore p$

Let p = it rains, q = the grass is wet
If it rains, then the grass will be wet
The grass is wet
$\therefore$ It rained

The grass could be wet from another source

Converse Error – Truth Table

		Premises		Conclusion
p	q	$p \rightarrow q$	q	p
T	T	T	T	T
T	F	F	F	T
F	T	T	T	F ??
F	F	T	F	F

A critical row is row in the truth table where all the premises of an argument are true.

Check for the critical rows where the conclusion is false.

Even when both premises are true, the conclusion can be false.

Since we found at least one case where true premises lead to a false conclusion, the argument form is invalid.

Inverse Error

$p \rightarrow q$
$\neg p$

$\therefore \neg q$

Let p = it rains, q = the grass is wet

If it rains, then the grass will be wet

It is not raining
$\therefore$ The grass is not wet

The grass could be wet from another source

Inverse Error – Truth Table

		Premises		Conclusion
p	q	$p \rightarrow q$	~p	~q
T	T	T	F	F
T	F	F	F	T
F	T	T	T	F ??
F	F	T	T	T

Observe the critical rows where the conclusion is **false**.

This argument form is **invalid.**

Example

Is the following argument **valid** or **invalid**?

If your code has a syntax error, then it will not compile.

Your code does not have a syntax error.

Therefore, your code will compile.

Application of rules of Inference

Suppose we know these facts about a program:

1. x is not equal to 5 and x is greater than y.
2. If y is prime, then x equals 5.
3. If y is not prime, then z is odd.
4. If z is odd, then z is less than x.

Can we conclude that z is less than x?

Application of rules of Inference

Let's use propositional variables instead:

p: "x equals 5"
q: "x is greater than y"
r: is "y is prime"
s: "z is odd"
t: "z is less than x"

Application of rules of Inference

Formalize **the premises** as propositions:

(a) $\neg p \wedge q$

(b) $r \rightarrow p$

(c) $\neg r \rightarrow s$

(d) $s \rightarrow t$

The **conclusion** is the proposition t.

Application of rules of Inference

Show that the premises "The code does not pass all tests and the database backup is complete," "We will deploy to production only if the code passes all tests," "If we do not deploy to production, then we will deploy to staging," and "If we deploy to staging, then we will notify the team" lead to the conclusion:

"We will notify the team."

Proofs

Proof

A proof is a valid argument that establishes the truth of a mathematical statement.

Formal proof:
Every step is the application of just one rule

Informal proof:
Combine multiple steps in one step

Example

This chapter focuses on mathematical statements

e.g. If $5x + 2 = 27$, then $x = 5$

This is a mathematical statement

You could prove this by applying algebra

$5x = 27 - 2 = 25$

$x = 25 / 5 = 5$

Application to Computer Science and Engineering

We need to develop programs to solve mathematical problems

- Not just addition, subtraction, etc., but problems that require mathematical *methods*
- e.g. graphics, security analysis, modeling, data analysis

We need to determine if our programming approach can solve our problem!

Assumptions

The basic laws of algebra apply

The following properties of equality apply:

A = A

If A = B, then B = A

If A = B and B = C, then A = C

The set of all integers (**Z**) is ***closed*** under addition, subtraction, and multiplication

Even and Odd Integers

We are looking at integers (set Z)

An integer n is even iff n = twice some integer
Formally: n is even $\Leftrightarrow \exists\ k \in Z \mid n = 2k$

An integer n is odd iff n = twice some integer plus one
Formally: n is odd $\Leftrightarrow \exists\ k \in Z \mid n = 2k + 1$

Activity

a) Prove that -10 is even?

b) If x and y are integers, prove that 4x + 2y + 1 is odd

Formally: n is even $\Leftrightarrow \exists\ k \in Z \mid n = 2k$

-10 = 2 * -5

-5 is an integer

therefore -10 is even!

Activity

a) Prove that -10 is even?

b) **If x and y are integers, prove that 4x + 2y + 1 is odd**

Formally: n is odd $\Leftrightarrow \exists\ k \in Z \mid n = 2k + 1$

4x + 2y + 1 – how to transform?

1. **$4x + 2y + 1 = 2(2x + y) + 1$**
2. **$2x + y$ is an integer or** $\in Z$ because Z is closed under addition and multiplication.
3. Therefore that **$4x + 2y + 1$ is odd**

Prime and Composite Integers

An integer n is prime iff n>1 and for all positive integers r and s, if n = rs, then either r or s equals n

More formally: n is prime $\Leftrightarrow (n>1) \wedge (\forall\ r,s \in Z^{+}, (n=rs) \longrightarrow (r = 1 \wedge s=n) \vee (s=1 \wedge r=n)$

An integer n is composite iff n>1 and there exists some integers r,s (1<r<n, 1<s<n) where n =rs

More formally: n is composite $\Leftrightarrow (n>1) \wedge (\exists\ r,s \in Z^{+}, (1<r<n, 1<s<n) \mid n = rs)$

Activity

Prove that 2 is a prime number

Prove that 4 is a composite number

Proving Existential Statements

$\exists x \in D$ such that Q(x) is true **iff** Q(x) is true for **at least one x** in D.

One way to prove this is to find an x in D that makes Q(x) true.

Another way is to give a set of directions for finding such an x.

Both of these methods are called **constructive proofs of existence**.

Example

Prove: ∃ even integer n that can be written as a sum of two composite numbers

We need to find **at least one** n where this is true.

Composite numbers are 4,6,8,9,10,12…

4 + 6 = 10, let n = 10. So this is true.

Disproving Universal Statements by Counterexample

To disprove a statement means to show that it is **false**.

Disprove: $\forall$ x in D, if P(x) then Q(x)

Showing that this statement is **false** is equivalent to showing that its negation is **true**.

The negation of the statement is existential: $\exists$ x in D such that **P(x) and not Q(x)**.

Example

Given the following statement:

$\forall\ x,y \in \mathbf{R}, (x^2 = y^2) \rightarrow (x = y)$

Find a **counterexample** to disprove this statement

$\exists\ x,y \in \mathbf{R} \mid (x^2 = y^2) \wedge \sim(x = y)$

So let $x = -2$, $y = 2$

$x^2 = y^2 \quad (-2)^2 = 4,\ 2^2 = 4$, so $4 = 4$

But, $-2 \neq 2$, so $\sim(x = y)$

Activity

Disprove the following statement by giving a counterexample:

"For all integers n, if n is odd then (n–1)/2 is odd."

Proving Universal Statements

How to prove a universal statement?

The majority of mathematical statements to be proved are universal.

$\forall x \in D$, if $P(x)$ then $Q(x)$

When **D is finite** or when only **a finite number of elements satisfy P(x)**, such a statement can be proved by the **method of exhaustion**.

Method of Exhaustion

$\forall n \in \mathbf{Z}$, if n is even and $\mathbf{4 \leq n \leq 16}$, then n can be written as a sum of two prime numbers. Prime numbers are 2,3,5,7,11,13

n	sum of primes
4	2+2
6	3+3
8	3+5
12	5+7
14	7+7
16	5+11

Application to Computer Science

We could **“exhaustively”** run a program to test all cases.

For large data sets, it would take a “long” time (many years even with the fastest computer)

We need a shortcut to verify the operation of our program.

Proving Universal Statements

By using the method of generalizing from the generic particular, we can then use the method of **direct proof** to prove *universal* statements

Common Proof Techniques

1. **Direct Proof** – Assume $P(x)$, show $Q(x)$ follows directly
2. **Proof by Contrapositive** – Prove $\neg Q(x) \rightarrow \neg P(x)$ instead
3. **Proof by Contradiction** – Assume $\neg Q(x)$ and derive a contradiction
4. **Proof by Cases** – Break into exhaustive cases
5. **Mathematical Induction** – Prove for base case and inductive step

Method of Direct Proof

Step 1: Identify the Universal Statement

- Express as: $\forall x \in D, P(x) \rightarrow Q(x)$
- Identify the domain D
- Identify hypothesis P(x)
- Identify conclusion Q(x)

Step 2: Use Method of Generalizing from Generic Particular

- Start: "Suppose x is an arbitrary element of D"
- State: "Assume P(x) is true"
- This makes x representative of ALL elements satisfying P(x)

Method of Direct Proof

Step 3: Apply Definitions

- Unpack the hypothesis P(x) using its formal definition
- Write out what P(x) means mathematically

Step 4: Use Logical Deduction

- Apply algebraic manipulation
- Use known theorems and properties
- Apply rules of inference
- Perform valid substitutions

Step 5: Arrive at Conclusion

- Show that Q(x) follows from your deductions
- State conclusion clearly: "Therefore, Q(x) is true"

Direct Proof Template

Theorem: $\forall x \in D$, if $P(x)$ then $Q(x)$

Proof:

Let x be an arbitrary element of D.
Assume $P(x)$ is true.

[By definition of P, we have...]
[Specific mathematical statements about x]

[Use algebra/logic to deduce...]
[Step-by-step deductions]

[Therefore, by definition of Q...]
Therefore, $Q(x)$ is true.

Since x was arbitrary, the statement holds for all $x \in D$.

Direct Proof – Key Components

What makes a direct proof work?

- Arbitrary element – Don't pick a specific value; use a generic variable
- Formal definitions – Always expand definitions (even, odd, prime, etc.)
- Logical chain – Each step must follow from previous steps
- Valid algebra – Only use legal algebraic operations
- Clear conclusion – Explicitly show you've proven Q(x)

Direct Proof Example

Prove the following statement:

"The sum of any two odd integers is even"

Proof: Suppose m and n are any [particular but arbitrarily chosen] odd integers. [We must show that m + n is even]

By definition of odd, there exist integers r and s such that **m = 2r+1** and **n = 2s+1**

m+n=2r+1+2s+1=2r+2s+2=2(r+s+1)

p = r + s + 1 is an integer as Z is closed under addition, r,s,1 are integers

2p is an integer by definition

Direct Proof Example

by substitution

$m + n = (2r+1) + (2s + 1)$

$m + n = 2r + 2s + 2$

$m + n = 2(r + s + 1)$

$u = r + s + 1$ is an integer

Activity

Prove the following statement:

"For all integers m, if m is even then 3m+5 is odd"

Direct Proof Example

Theorem: The sum of any two even integers is even.

Formal Statement: $\forall m, n \in \mathbb{Z}$, if m is even and n is even, then m+n is even

Let m and n be arbitrary integers. ← Generic particular
Assume m and n are both even. ← Hypothesis

By definition of even: ← Apply definition
$\exists k \in \mathbb{Z}$ such that $m = 2k$
$\exists j \in \mathbb{Z}$ such that $n = 2j$

Then: $m + n = 2k + 2j$ ← Substitution
$= 2(k + j)$ ← Algebra

Since $k + j$ is an integer, ← Closure property
$m + n = 2(k+j)$ is even by definition. ← Apply definition

Therefore, m + n is even.

Proof by Contrapositive

Alternative approach to proving $P \rightarrow Q$

Instead of proving $P \rightarrow Q$ directly, prove its contrapositive: $\neg Q \rightarrow \neg P$

Use this!

When $\neg Q \rightarrow \neg P$ is easier to prove than $P \rightarrow Q$

When the conclusion Q is a negative statement

When assuming $\neg Q$ gives you more concrete information to work with

Proof by Contrapositive Template

To prove $P \rightarrow Q$

1. Assume $\neg Q$
2. Show that $\neg P$ must follow

Proof by Contrapositive Example

Theorem: If n^2 is even, then n is even (where $n \in \mathbb{Z}$)

Why use the contrapositive?

Direct proof is harder:

- Assuming n^2 is even doesn't give us much to work with.
- But assuming n is odd gives us $n = 2k+1$, which is very useful!

Proof by Contrapositive Example

Prove that If n^2 is even, then n is even (where $n \in \mathbb{Z}$)

Contrapositive: If n is odd, then n^2 is odd

Proof:
Assume n is odd. ← Assume ¬Q (negation of conclusion)

By definition of odd, $\exists k \in \mathbb{Z}$ such that $n = 2k + 1$.

Then: $n^2 = (2k + 1)^2$ ← Substitution
$= 4k^2 + 4k + 1$ ← Expand
$= 2(2k^2 + 2k) + 1$ ← Factor out 2

Let $m = 2k^2 + 2k$. Since $k \in \mathbb{Z}$, we have $m \in \mathbb{Z}$.
Therefore, $n^2 = 2m + 1$, which means n^2 is odd by definition.

Since we proved $\neg Q \rightarrow \neg P$, the contrapositive $P \rightarrow Q$ is also true. ← Logical equivalence
Therefore, if n^2 is even, then n is even.

Proof by Contradiction

Assuming the opposite leads to the impossible.

Prove a statement is true, assume it's false and show this leads to a contradiction.

1. Assume the statement you want to prove is false (assume ¬Q)
2. Use logical deduction to derive a contradiction (something that's obviously false)
3. Since assuming ¬Q led to a contradiction, ¬Q must be false
4. Therefore, Q must be true

Proof by Contradiction

When to use?

- Proving uniqueness or existence
- When direct proof seems difficult
- Proving irrationality
- Proving infinitude of sets

Common contradictions

- $0 = 1$ or other false equations
- $n \neq n$ for some value n
- A number is both even and odd
- Something is both true and false

Proof by Contradiction – Example

Theorem: $\sqrt{2}$ is irrational
Note: A number is rational if it can be written as p/q where $p, q \in \mathbb{Z}$, $q \neq 0$, and p/q is in lowest terms ($\gcd(p,q) = 1$)

Proof:
Assume, for the sake of contradiction, that $\sqrt{2}$ is rational. ← Assume $\neg Q$
Then $\exists$ integers p, q with $q \neq 0$ and $\gcd(p,q) = 1$ such that $\sqrt{2} = p/q$.

Squaring both sides: $2 = p^2/q^2$
Therefore: $p^2 = 2q^2$ ← p^2 is even

Since p^2 is even, p must be even (by contrapositive from earlier!).
So $\exists\, k \in \mathbb{Z}$ such that $p = 2k$.

Substituting: $(2k)^2 = 2q^2$
$4k^2 = 2q^2$
$2k^2 = q^2$ ← q^2 is even
Since q^2 is even, q must be even.

But if both p and q are even, then $\gcd(p,q) \geq 2$, which **contradicts** our assumption that **$\gcd(p,q) = 1$!**
Therefore, our assumption that $\sqrt{2}$ is rational must be false. **Thus, $\sqrt{2}$ is irrational.**